University Roll No. CS B.Tech 2nd year

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Yr. (Odd Sem.), Term I Examination, 2016-17

Subject: English for Professional Purposes-I Subject Code: AHE 3083
Time: 50 Minutes Max. Marks: 15

Notes:

- Answer all the questions from Section A, any 3 out of 4 from Section B, and any 2 out of 3 from Section C.
- Answers should be to-the-point, and word limit for informative questions from Section B (approx 60 words) and Section C (150 words) should strictly be maintained.

Section A 0.5 x 6=3

1. Write one line definition of communication.
2. A doctor talking to a patient during a medical check-up is an example of **Verbal** type of communication. (True/False)
3. The VC talking to Pro-VC in GLA University is an example upward mode of communication. (True/False)
4. The first slide in Power point presentation must have at least five different colours. (True/False)
5. There is a boss who always talks to his employees very rudely. Which of the 7 'C's/features of good communication does he fail to follow?
6. Sangeeta can't talk in Hindi and John can't talk in English. Which barrier these two may face if they have to talk to each other?

Section B 2x3=6

1. **a).** Go through the following discussion:

   Shivangi: I am very hungry. I need something to eat immediately.
   Vatsala: I am sorry but I have nothing to offer you at this odd time of the night. You should not expect me to provide you with food all the time. I understand that you didn't eat anything in the evening; but how fair is it to expect me to feed you all the time, when you never pay anything to me. I really feel bad that you don't have even an iota of common sense.
   Shivangi: I am sorry.
   Which of the 7 'C's, do the speech of Vatsala, lack here? Give reasoned answer.

   b). Prevalence of English language in India has helped it a great deal in numerous ways, comment.

2. 'Lack of self-perception' is an important psychological barrier.' Elucidate.
3. Write any two advantages and two disadvantages of power point presentation over the traditional presentation methods.

4. Analyze the following telephonic conversation:

Smith:	Hello!
Jenny:	Hello!
Smith:	Madam! Could you please see me in my office? I have something important to talk to you.
Jenny:	Hello....Hellooooooo. Heloo sir! I am sorry. Please come again.
Smith:	I need to see you in my office. Could you please come?
Jenny:	I am sorry. I am not able to get you correctly. Do you want me to come to your office?
Smith:	Hello....Helloooo. Helooo....Please wait, let me send you a text message. Seems, there is some network issue.

Analyse the barrier in the above discussion that stopped the communication from going through.

b). Confidence and nervousness are not static but depend largely on situation, put your view in agreement or disagreement.

Section C 3x2=6

1. Explain the process of communication with the help of a flowchart with reference to encoding, decoding, medium and feedback.
2. Make the first two slides for TATA's proposal to launch a new car in December 2016. You should at least include features of the car, advantages over existing cars, and target customer information.
3. Write notes on any one of the following:
   a. Connotative and denotative language
   b. Organisational barrier

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University Roll No. .....

B.Tech. II Year, III Semester, I[st] Mid Term Examination (2016-17)

MBA-5031: Industrial Economics

Time: 90 Mins. **Total Marks: 20**

Section-A

Attempt All Questions **[5×1 = 5 Marks]**

I. Define Industrial Economics.
II. What the term 'Ceteris Paribus' reflects in Law of demand?
III. What do you mean by Income Effect and Substitution Effect?
IV. What is the concept of Timeless Economy?
V. Mention the basic characteristics of resources giving rise to economic problems.

Section- B

Attempt any THREE Questions **[3×2 = 6 Marks]**

I. Economics is an art or a science? Discuss.
II. What is Demand Function? Discuss Law of Demand along with assumptions and exceptions.
III. What are various methodologies of economics? Discuss along with the help of suitable examples.
IV. A person purchases 100 units of product X when its prices are Rs.15per unit and related product Y is available at Rs. 18per unit. With a change in prices of product Y to Rs.16per units, demand for product X declines to 90units. Calculate Cross Elasticity of Demand.

Section- C

Attempt any THREE Questions **[3×3 = 9 Marks]**

I. How Industrial Economics helps a Business firm and Government? Discuss.
II. Discuss the concept of Elasticity of Demand along with its types. Write a brief note on methods of calculation also.
III. "Ceteris Paribus, with the successive consumption of units of a specific commodity the marginal utility diminishes constantly". Comment and discuss the concept with suitable example. What are its exceptions?
IV. Discuss the relationship between science, engineering, technology and economics.

Printed Pages-2 **University Roll No.....**

B. Tech. II Year III Semester,

First Term Examination, 2016-17

CSE3001: Data Structures and Algorithms

Time: 01 ½ Hours **Total Marks: 20**

Note:

1. Attempt **all** questions from Section-A, **any three** from Section-B and **any three** from Section-C.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Starting from an empty doubly linked list, the following operations are performed, in order: addFirst(A), addFirst(B), addLast(C), addLast(D), insertBefore (2,E), insertAfter(3, F), remove(2), where indices start at 0 and A, B, etc, are instances of the Node interface. Draw the list that results after those operations. Draw only the final result.

II. What is Big O notation?

III. Convert the following arithmetic infix expression into postfix expression

A+ (B*C-(D/E^F)*G)*H

IV. Explain stack as an ADT.

V. What is a Dequeue? Mention one application of the same.

Section B

Note: Attempt any Three Questions (2x3=6)

I. Mention at least two usage of Circular queue. Write the algorithm for deletion from a circular queue implemented using an array.

II. Mention at least two usage of Stack. Write the algorithm for deletion from a stack implemented using a linked list.

III. Write an algorithm for deletion of the last node in a doubly linked list.

IV. Write the algorithm to evaluate a postfix expression.

Section C

Note: Attempt any Three Questions (3x3=9)

I. Here is an infix expression: 4+3*(6*3-12). Suppose that we are using the usual Stack algorithm to convert the expression from infix to postfix notation. What is the maximum number of symbols that will appear on the stack AT ONE TIME during the conversion of this expression? Support your answer with status of stack in various steps during the conversion.

II. Write an algorithm to search for nodes in linked list whose values are equal to 11. If the number of such occurrences are even in number then it should replace all such instances by value 100 else replace all such instances by value 99. Further, it should return the positions of these nodes, if found, otherwise it should return 0.

III. Write a C program to add two polynomials using linked list.

IV. Write a C program to implement a priority queue.

University Roll No | | | | | | | | | | |

First Mid Term Examination, Odd Semester 2016-17

B.Tech. (II Year)- III Semester

Subject:- Mathematics - III **Subject Code:- AHM-3101**

Time: 1 ½ Hours **Max. Marks: 20**

Section-A

Note: ***Attempt All Questions*** **1×5=5**

1. Form the partial differential equation from $z = f(x^2 - y^2)$.
2. Solve the partial differential equation $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = 0$
3. Solve the partial differential equation $(D - D')(D - 2D' - 3)^2 z = 0$
4. Solve the partial differential equation $zp + yq = x$.
5. Classify the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} - x\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial z}{\partial y} = 0$$ in second quadrant.

Section –B

Note: ***Attempt Any three Questions:*** **2×3=6**

1. Solve by method of separation of variables

$$\frac{\partial^2 u}{\partial x \partial t} = e^{-t}\cos x, \quad u(x, 0) = 0 \text{ and } \frac{\partial u}{\partial t} = 0 \text{ at } x = 0$$

2. Solve the partial differential equation

$$x(y-z)p + y(z-x)q = z(x-y)$$

3. Solve the partial differential equation

$$4\frac{\partial^2 z}{\partial x^2} - 4\frac{\partial^2 z}{\partial x\,\partial y} + \frac{\partial^2 z}{\partial y^2} = 16\,\log(x+2y)$$

4. Solve the partial differential equation

$$(D\,D' + a\,D + b\,D' + ab)\,z = e^{mx+ny}$$

Section – C

Note: ***Attempt Any three Questions*** : 3×3=9

1. Solve completely the equation $\frac{\partial^2 y}{\partial t^2} = c^2\frac{\partial^2 y}{\partial x^2}$, representing the vibrations of a string of length l, fixed at both ends, given that

$$y(0,t)=0, y(l,t)=0; y(x,0)=f(x) \text{ and } \frac{\partial}{\partial t}y(x,0) = 0,\ 0\le x\le l.$$

2. An insulated rod of length l, has its end A and B maintained at 0^0C and 100^0C respectively until study state condition prevail. If B is suddenly reduced to 0^0C and maintain at 0^0C, find the temperature at a distance x from A at time t.

3. Solve the partial differential equation

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x\,\partial y} - 2\frac{\partial^2 z}{\partial y^2} = (y-1)e^x$$

4. Solve the linear partial differential equation

$$(D^2 - DD' + D' - 1)\,z = \cos(x+2y) + e^y.$$

Printed Pages-2 **University Roll No.....**

First Term Examination, 2016-17
Course: B.Tech (CSE), Year: II, Semester: III
Computer Organization (CSE3002)

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. Find the 2's complement of $(101.1101)_{10}$.
II. Enlist the five building blocks of digital computer.
III. How many check bits are needed to detect single bit error in 512 bits data word (Show all Steps)?
IV. What is the value of base b if the decimal roots of equation $X^2-(10)_b X + (31)_b = 0$ are 8 and 5?
V. Express (-105) in IEEE 754, 32- bit floating point format.

Section B

Note: Attempt any Three Questions **(2x3=6)**

I. A digital computer has a common bus system of 64 registers each of 32 bits. The bus is constructed with a multiplexer.
   a) How many multiplexers are there in the bus?
   b) What size of multiplexer are needed?
II. Draw and explain Carry-Look-Ahead Adder with its advantages over other Adders.

III. Describe the block diagram of the hardware that implements the following register transfer language statement

If (P=1) then (R1←R2) else if (Q=1) then (R1←R3)

IV. Convert the following numbers from the given base to the other three bases listed in the table.

Decimal	Ternary	Octal	Hexadecimal
265			
	121		
		512	
			A3C7

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Perform multiplication between (+17) and (-6) by using Booth's multiplication algorithm and verify the result.

II. Draw block diagram of binary addition-subtraction algorithm for sign-magnitude representation. Derive an algorithm in flowchart form for adding and subtracting binary numbers.

III. Design an array multiplier that multiplies two 3-bit numbers. How many AND gates require to design it?

IV. What do you mean by Error Detection and Error Correction? Suppose an 11-bit data word stored in memory is 11011111011. Using Hamming algorithm, determine what check bits would be stored in memory with the data word?

[SET B]

Printed Page: 9 **University Roll No:**

First Term Exam (ODD SEMESTER)-2016

Ethics & Values: AHE 3001

Time: 1 ½ Hours **Max. Marks: 20**

Note: There are four passages in this paper. These passages are based on course discussion.

ONLY ONE OPTION IS CORRECT.

Carefully read the passages and answer the following questions. Carefully fill all the information in OMR sheet neatly in the blocks. No partial marking.

Passage-1 (1 X5 = 5 Marks)

In our lives, we come across several situations in which we evaluate actions and conduct of people. Some acts are approved and are called right or good. Other acts are condemned and are called wrong or evil. When we approve or disapprove of something, we are making a particular kind of judgement called moral judgement. Moral judgements have to do with the actions of human beings, insofar as questions of right and wrong are involved. Ethics is the study of principles underlying the moral judgements. The terms *morals* and *ethics* are closely related in original meaning. These terms are often used interchangeably. However in popular usage morality is more often used to denote the ideas of right and wrong, good or bad, that we have developed under the influence of the social, religious and cultural conditions in which we live. Ethics, on the other hand, is a systematic re-examination of the concepts of morality that are formed in our minds through critical reasoning. Hence it is called a second order study.

Within ethics, there are a number of areas of interest and study. First, there are various moral philosophical views or ethical theories which set out to explain why people act as they do and what are the principles by

which men live or should live. Then there is the wide range of human conduct ranging from the conduct of an individual to that of a group or a society or an institution which we examine in the light of those theories. Ethics seeks to establish a norm or standard for action so that men may choose their own actions with greater understanding and be able to evaluate critically the practices and institutions of society.

Questions:

1) Ethics involves:
1.Critical attitude
2.Scientific reasoning
3.Questioning morals of a society
4.Rationality.
a)3&4 b)1,2, 3& 4 c) Only 2 d) only 1& 2

2) Morality and Ethics develop:
1. The right kind of personality
2. The critical ability to deal with the life issues.
3. The ability to understand the physical nature.
4. Only rationality.
a) 1& 4 b) 1, 2 & 4, c) 1& 2 d) only 4

3) Ethics is called a second order study because
1.It is a critical study.
2.It examines morality.
3.It does not raise questions on morality.
4.It involves self interest only

a)1&4 b) 1&2 c) 3 d)1&4

4) Moral judgement involves
1.Rightness of action
2.Goodness of action
3.Necessity of an action
4.Art of an action
a)1, 2 & 3 b) Only 1, c) 2 & 3 d) only 4

5) Social and religious conventions
1.Shape our morality

B

2.Develop our scientific attitude
3.Help us develop an ethical viewpoint
4. Have to be questioned critically.

a)2&3 b) 1&2 c)2 d)1&4

Passage: 2 (1x 5 =5 marks)

Ethical Decision Making is composed of the ability to engage in ethical discourse with rationality, critical thinking, problem solving and decision making to analyze the life issues (problems of one's personal life, social life, and professional life). What could be the most challenging other than making a decision? Indeed, it's ethical decision making. Decision making is always a tough task to perform but deciding in a situation where the choices are neither completely right nor completely wrong; the decision making gets tougher. In decision making a person is examined in some pressing situations.

Critical thinking is a way we question our assumptions and prejudices and try to look at the problem with a fresh view. Problem solving is developed through these above mentioned abilities and is a kind of capacity to understand problems and arrive at the solutions. The ethical point of view is necessary for a decision maker. It may be the case that someone decides through certain reasons, however, does not think from an ethical point of view whether his/her decision would affect someone or not. This is the reason why an instrumental decision making differs from that of ethical decision making (hereafter, E.D.M.). In instrumental decision making process though we apply reasoning and critical thinking but the process aims to achieve desired results. The process is primarily concerned with non-ethical result oriented concerns. Cost-benefit analysis and bottom line approaches are some instances of instrumental decision making. In E.D.M. we consider not only the desired results but also the rightness and wrongness of our decisions. Ethical decision making requires giving up our narrow self interest and its concerns are value-based. An action that adversely affects the society or the environment or does not respect individual freedom and virtue is reconsidered. The ideal of People, Planet and Profit is applicable to all

decision making. It is to have the concern and care for the nature, humanity and wealth creation.

Questions:

6) Which of the following statements are correct about the Ethical Decision Making?

1. It is necessary for a decision maker.
2. It is an instrumental sort of decision making.
3. It is concerned with non-ethical concerns.
4. It presupposes rationality.

a) 2&3 b) Only1 c)1&4 d) Only 2

7) Where do we use the problem solving and decision making?

1. In professional life only.
2. In both professional and social life.
3. It can be used in personal life as well.
4. Only in professional and social life.

a)1&2 b)1,2&3 c)Only 2 d) Only 4

8) What do you understand by the value based ethical concerns?

1.The concern is about social responsibility.
2.They take into account individual freedom and virtue.
3.They are concerned with environmental safety and social welfare.
4. It's only concerned with cost and benefits.

a) Only 1 b) Only 3 c) 1,2 &3 d)Only 4

9) Which of the following statements are true about decision making?

1. It requires both rationality and critical thinking.
2. Decision making gets tougher when choices are neither completely right not fully wrong.
3. Decision making is more challenging than ethical decision making.
4. It only requires problem solving.

a) Only 4 b) 1&2 c)Only 3 d) 3 &4

10) What is instrumental decision making?

1.The process is based on non-ethical concerns.
2.Cost-benefit analysis and bottom line approaches are some example of instrumental decision making.

3.In this process though we apply rationality and critical thinking but it aims to achieve desired results.

4. It requires thinking beyond our narrow self interest

a) Only 1 b)Only 3 c) 1,2 &3 d) 1 &4

Passage: 3 (1x 5 =5 marks)

Surrogacy is the practice by which another woman carries and gives birth to a baby for a couple who want to have a child. The woman who agrees to carry the child and give birth to a baby is called a Surrogate Mother. If the surrogate receives money for the surrogacy, the arrangement is called Commercial surrogacy. In India Commercial Surrogacy has been legal since 2002 and many foreigners have been coming to India for surrogacy. However a new bill has been recently proposed by the government which seeks to ban this practice. The Government has said that it does not support commercial surrogacy and will certainly not allow surrogacy for foreigners. Only Indian Couples who are childless after having been married for five years can opt for this practice. In addition, the surrogate mother will have to be close relative of the couple.

Many ethical issues have been raised in light of the controversy surrounding this bill. The supporters of the ban say that due to surrogacy, exploitation of poor women is taking place. They also argue that commoditisation of motherhood and babies are going on because of this practice. Some further say that certain sacred notions like that of motherhood, human life were being devalued because of surrogacy. On the other hand, the opponents of the ban say that surrogacy is a noble practice which benefits parties, the childless couple and the surrogate. They further argue that it is a woman's right to make contracts regarding the use of her body. They believe in the principle that an individual is the ultimate owner of his or her body, not the government. The opponents also question the right of a government to force childless couples to choose surrogates among relatives.

Questions:

11) A supporter of surrogacy favours the practice because

B

1. It benefits both parties, the surrogate and the couple
2. It is a noble act
3. It values motherhood in terms of money
4. It respects women's freedom and rights

a) Only 2 b)1,2 & 4 c)Only 3 d) Only 1

12) An opponent of Surrogacy opposes it because

1. It devalues motherhood
2. Babies born out of surrogacy are unhealthy
3. It leads to exploitation of women
4. It commercialises human life

a) Only 3 b) Only 1 c) 1,3 & 4 d) 1&2

13) A surrogate mother

1. Is one who adopts someone's child
2. Agrees to carry the child of another couple and give birth to their baby
3. Is someone who leaves her child in the orphanage
4. Is one who is willing to carry a child of another couple only if she gets paid.

a) 1&3 b) Only 4 c)Only 3 d) Only 2

14) The government proposes to

1. Ban commercial surrogacy completely
2. Ban only Indian surrogacy
3. Allow foreigner surrogacy only
4. .Allow only Indians under some conditions

a)Only 2 b)2 & 3 c) 1&4 d) Only 3

15) As per the proposed new law, a couple's surrogate can be

1. a stranger only
2. a foreigner
3. a relative
4. an Indian only

a) 1&4 b)Only2 c) Only1 d) 3&4

Passage: 4 (1x 5 =5 marks)

B

Trolley Problem is a hypothetical case in which we test one's ethical decision making. Hypothetical cases involve some imaginary situations in which the uncertainty regarding the outcome has been removed. This helps in bringing out our inner moral attitudes.

In this case as we all know there are three different versions that actually point out the variations in our decision making with the changes in the conditions. The case of trolley car also shows that our decision making is not simply based on reasons but also uses certain emotions at the root of which lie certain hidden reasons.

In the first version of the trolley car case: Suppose you are the driver of a trolley car hurtling down the track at sixty miles an hour. Up ahead you see six workers standing on the track, tools in hand. You try to stop, but you cannot. The brakes do not work. You feel desperate, because you know that if you crash into these six workers, they will all die. Suddenly you notice a side track off to the right. There is a worker on the track, too, but only one. You realize that you can turn the trolley car onto the side track, killing the one worker, but saving the six. How would you respond in this situation as a driver? The response of the majority is that that they would turn the trolley on to the side track with one person with the reason that it would spare six lives. The principle behind the decision seems to be that it is better to kill one person than kill six people, given the circumstances.

Let's look at the second version now: This time the Trolley car is automatic with no driver and is racing down a track. There is no side track. There is a bridge above the track on which you are standing as an onlooker. Down the track comes a trolley and you see that at the end of the track are six workers. Once again brakes do not work and trolley car is about to crash into those six men. You feel helpless to prevent this disaster. You think of jumping on the track but finding yourself too thin to stop the trolley you change your mind. However, you notice alongside you a very fat man just leaning onto the bridge. You reasonably think that if you give him a gentle push he will fall on the track before the trolley car and the track will be obstructed somehow. Although the fat man will die, he would save the life of six men. The response of the

majority in this version however is that they would not give a gentle push to fatty man to save six people on the track. Why is the response this time different when the result is same as in the first version- six lives will be saved at the cost of one person's life.

If we look at the difference in the circumstances compared to the previous one, we see that in this version the onlooker is not directly involved in the situation like the driver in the first case. But, why is the majority not mindful of the better consequences despite the result being the same-six lives saved at the cost of one.

This leads us to believe that people time do not keep in mind consequences only when evaluating a situation. It seems like we only go for better consequences if the nature of the action itself is not objectionable. In this case, we find that throwing an innocent fat man is a cruel act even though it leads to a better result.

Questions:

16.Why do you think the majority decline to push the fat man?

1.It means direct involvement.

2.Majority think that the decision should be left to fate.

3.It is wrong to take the life of the other.

4.The action is wrong in itself.

a) 1&2 b)Only 2 c)1,3&4 d) Only 1

17) Which of the following statements are **not true** about the second version of trolley car case?

1.Majority would like to jump before the trolley car.

2.Majority believe that one person can be killed to save six.

3.Majority would decline to push the fat man.

4.Majority would decline to save six men.

a) Only1 b) 1&2 c) 3 &4 d) Only 4

18) How do the three versions of trolley car case differ from one another?

1.On the grounds of reasons.

2.On the grounds of different conditions.

3.On the grounds of intentions.

4.On the grounds of better consequences

a) Only 3 b) Only 1 c) 1,2 &3 d)Only 4

19) What are the learning from the Run way trolley car case?

1. Our Ordinary rationality may be wrong
2.Critical thinking offers better reasons.
3.Critical thinking should not be done before any decision making.
4. Rationality does not require critical thinking.

a)1&2 b)Only 3 c)3&4 d) Only 4

20) Which of the following statements are true about this case?

1.It is a hypothetical case.
2.Majority would turn the trolley in second version.
3.Majority would not push the fat man in second version.
4.Majority would turn the trolley in the first version.

a) Only 4 b) 1,3&4 c) only 2 d) Only 3

First Term Examination, 2016-17
Course: B.Tech. (CSE), Year:II, Semester:III
Database Management System (CSE 3003)

Uni. Roll No: _____

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Section A

Note: Attempt All Questions **(1x5=5)**

I. What is a NULL value?

II. What are the major disadvantages of a database system?

III. Consider the following relational schema pertaining to a student's database: Students (*rollno*,name, address)

Enroll(*rollno, courseno*, coursename)

Where primary keys are shown in italics. The number of tuples in the student and Enroll tables are 120 and 8 respectively. What are the maximum and minimum number of tuples that can be present in (Student *Enroll), where * denotes natural join?

IV. Differentiate between weak and strong entity set.

V. What is a data model? List the types of data model.

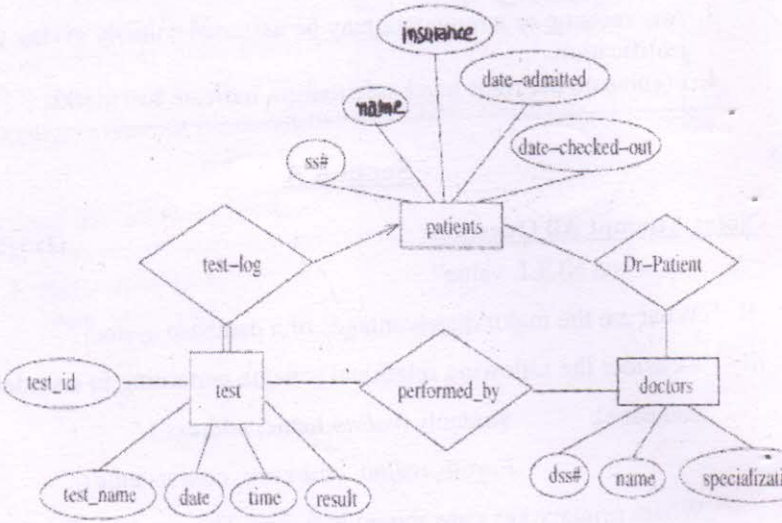
Section B

Note: Attempt any Three Questions **(2x3=6)**

I. Define the concept of aggregation. Give two examples of where this concept is useful.

II. Explain the types of participation constraints with an example?

III. Construct appropriate tables for the following ER Diagram.



E-R diagram for a hospital.

IV. What is relational calculus? What are its types? Write an expression in tuple relational calculus using tuple variable, conditions and WFFs.

Section C

Note: Attempt any Three Questions **(3x3=9)**

I. Consider the following relational Schema and give the answers of following in relational algebra:

Dealer(Dealer_no, Dealer_name, Address)

Part(Part_no, Part_Name, Color)

Assigned_To(Dealer_no, Part_no, cost)

a. Find names of dealers that supply red parts.
b. Find names of parts supplied by dealers lived in New Delhi.
c. Find names of dealers that supply both red and yellow parts.
d. Find names of dealers who supply all parts.

II. Consider the two tables T1 and T2 shown in figure. Show the results of the following operations:

Table T1

P	Q	R
10	a	5
15	b	8
25	a	6

Table T2

A	B	C
10	b	6
25	c	3
10	b	5

a. $T1 \bowtie_{T1.P=T2.A} T2$
b. $T1 \cap T2$
c. $T1 - T2$
d. $\pi_{P,}(T1 \bowtie_{T1.P<T2.A} T2)$
e. $T1 \overline{\bowtie} T2$

III. Suppose you are given the following requirements for a simple database for the National Hockey League (NHL):

- the NHL has many teams,
- each team has a name, a city, a coach, a captain, and a set of players,
- each player belongs to only one team,
- each player has a name, a position (such as left wing or goalie), a skill level, and a set of injury records,
- a team captain is also a player,
- a game is played between two teams (referred to as host_team and guest_team) and has a date (such as May 11th, 2014) and a score (such as 4 to 2).

Construct a concise ER diagram for the NHL database.

IV. Explain the following term:

a. Data redundancy and consistency
b. Referential Integrity
c. Data Atomicity

GLA University

SET-A (All)

First Term Examination (Soft Skills)-2015-16

Course- B.Tech **Year- II** **Branch-All**

Instructions :

1. There are 2 sections in the paper.
2. Attempt all questions. The maximum marks for the test are 20.
3. The total time allocated is 60 minutes.
4. There will be no negative marking. Also, no marks will be awarded if more than one answer is marked.
5. The OMR form is printed on a special paper and the information given by the candidates will be scanned. Thus, the candidates are required to fill the forms carefully. Mutilated and damaged forms will be rejected by the scanner.
6. The candidates should ensure that the form is not folded, crumpled or mutilated in any manner. The candidates should not mark anything anywhere except wherever required. The black marks on the margin of the OMR form are important. The candidates should not tamper it with marker.
7. Only black
8. & blue pen should be used to fill forms.
9. Each branch is identified with a specific subject code which should be written in subject column of OMR sheet
10. Write your University roll number in the roll no column of OMR sheet.

Name of Student:-.....

Branch:-.....

Section-.....

Uni. Roll No:.....

S.N	Branch	Code
1	CS	1500
2	CE	1000
3	ECE	1300
4	EN	1400
5	EE	1100
6	ME	1200

Signature of Invigilator-.....

Section-1 (Quantitative Aptitude)-No of question-20, marks-10

1. $(\sqrt{6561}) / (\sqrt[3]{729}) = ?$
(a) 81 (b) 27 (c) 9 (d) 3

2. $\sqrt{12321/37} = ?$
(a) 2.0 (b) 2.5 (c) 3.0 (d) None of these

3. If 5967X13 is divisible by 3. Find X.
(a) 2 (b) 5 (c) 8 (d) All of the above

4. If 4832X18 is divisible by 11. Find X.
(a) 7 (b) 8 (c) 9 (d) 1

5. If 12834785X is divisible by 4. Find X.
(a) 2 (b) 6 (c) both (a) and (b) (d) none

6. Mr. Amitab Bachhan is worried about the balance of his monthly budget. The price of petrol has increased by 50%. By what percent should he reduce the consumption of petrol so that he is able to balance his budget?
(a) 33.33 (b) 28.56 (c) 25 (d) 14.28

7. A number is mistakenly divided by 5 instead of being multiplied by 5. Find the percentage change in the result due to this mistake.
(a) 96% (b) 95% (c) 2400% (d) 200%

8. 30% of a number when subtracted from 91, gives the number itself. Find the number.
(a) 60 (b) 65 (c) 70 (d) 75

9. The population of the village of Chauma is 10,000 at this moment. It increases by 10% in the first year. However, in the second year, due to immigration, the population drops by 5%. Find the population at the end of the third year if in the third year the population increases by 20%.
(a) 12,340 (b) 12,540 (c) 1,27,540 (d) 12,340

10. At an election, the candidate who got 56% of the votes cast won by 144 votes. Find the total number of voters on the voting list if 80% people cast their vote and there were no invalid votes.
(a) 360 (b) 720 (c) 1800 (d) 1500

11. In an examination, Uday Chopra obtained 20% more than Govinda but 10% less than Chunky Pandey. If the marks obtained by Govinda are 1080, find the percentage marks obtained by Chunky Pandey if the full marks is 2000.
(a) 86.66% (b) 72% (c) 78.33% (d) 77.77%

12. Vicky's salary is 75% more than Ashu's. Vicky got a raise of 40% on his salary while Ashu got a raise of 25% on his salary. By what percent is Vicky's salary more than Ashu's?
(a) 96% (b) 51.1% (c) 90% (d) 52.1%

13. The population of a village is 1,00,000. Increase rate per annum is 10%. Find the population at the starting of the fourth year.
(a) 1,33,100 (b) 1,21,000 (c) 1,33,000 (d) None of these

14. 4/5th of the voters in Ramgarh promised to vote for Jai and the rest promised to vote for Veeru. Of these voters, 10% of the voters who had promised to vote for Jai, did not vote on the election day, while 20% of the voters who had promised to vote for Veeru did not vote on the election day. What is the total no. of votes polled if Jai got 216 votes?
(a) 200 (b) 300 (c) 264 (d) 100

15. The entrance ticket at the Minerva theatre in Mumbai is worth Rs. 250. When the price of the ticket was lowered, the sale of tickets increased by 50%, while the collections recorded a decrease of 17.5%. Find the deduction in the ticket price.
(a) 150 (b) 112.5 (c) 105 (d) 120

Directions for Questions 16 to 19:
Read the following and answer the questions that follow.
Two friends Shayam and Kailash own two versions of a car. Shayam owns the diesel version of the car, while kailash owns the petrol version. Kailash's car gives an average that is 20% higher than Shayam's (in terms of litres per kilometre). It is known that the price of petrol costs 60% of its price more than diesel's price.

16. The ratio of the cost per kilometre of kailash's car to shayam's car is
(a) 3 : 1 (b) 1 : 3 (c) 1.92 : 1 (d) 2 : 1

17. If shayam's car gives an average of 20 km per litre, then the difference in the cost of travel per kilometre between the two cars is
(a) 4.3 (b) 3.5 (c) 2.5 (d) Cannot be determined

18. For Question 17, the ratio of the cost per kilometre of shayam's travel to kailash's travel is
(a) 3 : 1 (b) 1 : 3 (c) 1 : 1.92 (d) 2 : 1

19. If diesel costs Rs. 12.5 per litre, then the difference in the cost of travel per kilometre between Kailash and shayam's is (assume an average of 20 km per litre for shayam's car and also assume that the price of petrol is 50% of its own price more than diesel's price)
(a) 1.75 (b) 0.875 (c) 1.25 (d) 1.125

20. If 25% of a number is 75, then 45% of that number is:
(a) 145 (b) 125 (c) 150 (d) 135

Section-2 (Reasoning Aptitude)-No of question-20, marks-10

Directions for 21 to 25:
(I) Agra is north of Bombay and east of Faridabad.
(II) Bombay is north of Faridabad and west of Calcutta.
(III) Delhi is south of Erode and west of Faridabad.
(IV) Erode is south of Faridabad and east of Agra.
(V) Calcutta is south of Erode and west of Delhi.

21. Which of the towns mentioned is farthest to the northwest?
(a) Agra (b) Bombay (c) Calcutta (d) Erode

22. Which of the following is/are to the north-east of Calcutta?
I. Agra
II. Bombay
III. Erode
IV. Delhi
(a) I only (b) II only (c) III only (d) I and III

23. Which of the following towns must be situated both south and west of at least one other town?
(a) Agra Only (b) Erode and Faridabad (c) Delhi and Agra (d) Bombay and Faridabad

24. Which statement is not having sufficient information to make it true?
(a) Calcutta is north of Delhi. (b) Erode is north of Delhi.
(c) Agra is east of Bombay. (d) Bombay is north of Faridabad.

25. Which of the following is to the south-west of Bombay ?
(a) Agra (b) Erode (c) Calcutta (d) None of these

Directions for 26 to 28:
Read the following information carefully and answer the questions given below it.
1. Six flats on a floor in two rows facing north and south are allotted to A, B, C, D, E and F.
2. There are three flats in each row and flats are facing to each other.
3. B gets a south facing flat and is at the end of the west side.
4. D's and F's flats are farthest and diagonally opposite.
5. E, next to F and gets a North facing flat.

26. Which of the following combination gets North facing flats?
(a) AEF (b) ECF (c) Data Insufficient (d) Either (a) or (b)

27. Which of the following are at extreme ends of the rows?
(a) AFB (b) CBD (c) BDA (d) FDB

28. Which of the following two persons flats are exactly opposite to each other?
(a) A&E (b) F&B (c) D&C (d) None of these

29. Four aeroplanes of Air Force viz., A, B, C and D started for a demonstration flight towards east. After flying 50 km planes A and D flew towards right, planes B and C flew towards left. After 50 km, planes B and C flew towards their left, planes A and D flew towards their left. In which directions are the aeroplanes A, B, D, C respectively flying now?

(a) North, South, East, West (b) South, North, West, East

(c) East, West, East, West (d) West, East, West, East

30. If South-East becomes East, North-East becomes North and so on, what will West become?
(a) North-East (b) South-West (c) South-East (d) North-West

31. One morning after sunrise, Rakesh was facing a pole. The shadow of the pole fell exactly to his left. Which direction was he is facing?
(a) North (b) East (c) West (d) South

32. Sohan walks 10 km toward North. From there, he walks 6 km towards South. Then, he walks 3 km towards East. How far and in which direction is he with reference to his starting point?
(a) 2 km South-East (b) 5 km South-East
(c) 5 km North-East (d) 5 km West

33. A girl leaves from her home .She first moves 30 m in North-West direction and then 30 m in South-West direction. Next, she walks 30 m in South-East direction .Finally ,she turns towards her house .In which direction is she moving?
(a) East (b) North-East (c) North (d) South-East

34. Dheeraj walks 10 m towards the East .He takes right turn and walks 20 m and then he takes left turn and moves 10 m. After that, he again takes right turn and then left turn and moves 20 m and 10 m respectively and so on. After his 7th turn he finished his journey. Find out the distance travelled by Dheeraj after 2nd right turn and before 3rd left turn?
(a) 20 m (b) 40 m (c) 50 m (d) 30 m

Questions (35-37):
Seven villages P, Q, R, S, T, U and V are situated as follows:
T is 10 km to the west of Q. U is 10 km to the north of P.
R is 5 km to the West of P. S is 10 km to the south of V.
V is 10 km to the east of R. S is the Exactly in the middle of Q and T.

35. P is in the middle of
(a) T&R (b) U&T (c) U&V (d) V&R

36. Which two villages are the farthest from one another?
(a) S&R (b) U&T (c) U&Q (d) V&T

37. How far away is V with respect to U ?
(a) 10 km (b) $5\sqrt{5}$ km (c) 20 km (d) $\sqrt{13}$ km

38. A clock reads 4.30. If the minute hand point East, In what direction will the hour hand Point?
(a) North (b) North-West (c) South-East (d) North-East

39. One day afternoon, Megha and Riya are taking to each other at Connaught Place in Delhi. Riya's shadow fells exactly to the front of Megha. In which direction is Megha facing?
(a) East (b) West (c) South (d) North

40. A boy went 100 m in the East to look for his friend, then he turned right and went 30 m. After this he turned right and after going 40 m he reached to his uncle's house. His friend was not there. From there he went 110 m to the north and met his friend. In which direction did he meet his friend with respect to starting point?
(a) North-East (b) North-West (c) North (d) South-West

SET-A University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Odd Semester), Term I Examination, 2016-17

Subject: English for Professional Purposes–II Subject Code: AHE 3083
Time: 40 Minutes Max. Marks: 20

VERBAL APTITUDE

Identify the part of speech of the underlined word in Q. 1-10.

1. None <u>but</u> Vijay can tackle this problem.
   A. Preposition B. Conjunction C. Adjective
   D. Connector E. None of these
2. Only <u>one</u> person is allowed in the I.C.U.
   A. Preposition B. Pronoun C. Adjective
   D. Noun E. None of these
3. A bird has to feed its <u>young</u>.
   A. Pronoun B. Adjective C. Adverb
   D. Noun E. None of these
4. It's impossible to reach by Friday <u>for</u> I have a wedding to attend here on the same day.
   A. Adjective B. Preposition C. Verb
   D. Conjunction E. None of these
5. <u>Before</u> I became a teacher, I worked as a columnist.
   A. Preposition B. Adjective C. Adverb
   D. Pronoun E. None of these
6. John is tall <u>but</u> Rosie is short.
   A. Preposition B. Verb C. Adjective
   D. Conjunction E. None of these
7. Bushra is too <u>young</u> to be married off.
   A. Noun B. Pronoun C. Adjective
   D. Adverb E. None of these
8. She can dance <u>for</u> hours and hours.
   A. Preposition B. Conjunction C. Connector
   D. Adverb E. None of these
9. <u>Luckily</u>, I survived the accident.
   A. Adjective B. Adverb C. Noun
   D. Verb E. None of these
10. <u>One</u> should love one's neighbour.
    A. Noun B. Adjective C. Preposition
    D. Pronoun E. None of these

Identify the part that contains error (Q. 11-15).

11. King Lear, who (A) / was wandering (B) / about the fields near Dove, (C) / were found by some men (D) / in pitiable condition (E).
12. English is an official language (A) / or a joint official language (B) / in about fifty other countries, where it is use in addition to (C) / the indigenous first languages (D) / for a variety of public and personal functions (E).
13. The number of first-language(A) / speakers of English have been (B) /estimated at well over 300 million,(C) / of whom over 216 million (D) / live in the United States (E).
14. Once upon a time, (A) / there was a general manager (B) / in a downtown high-rise (C) / who have the final say over (D) / all new hires at her location (E).
15. There are just (A) / too many interviewing (B) / books and training tools (C) / available that teaches people (D) / the right things to say (E).

Identify the grammatically correct sentence (Q. 16-20)

16. A. A flock of sheep were grazing in the field.
    B. A flock of sheep have been grazing in the field.
    C. A flock of sheep was grazing in the field.
    D. A flock of sheep are grazing in the field.
    E. A flock of sheep graze in the field.
17. A. Lying in a poor bed, Robert Bruce look up at the ceiling and saw a spider.
    B. Lying in a poor bed, Robert Bruce looking up at the ceiling and saw a spider.
    C. Lying in a poor bed, Robert Bruce have looked up at the ceiling and saw a spider.
    D. Lying in a poor bed, Robert Bruce have been looked up at the ceiling and saw a spider.
    E. Lying in a poor bed, Robert Bruce looked up at the ceiling and saw a spider.
18. A. John, an officer in the revenue department, come here on Tuesday to meet his ailing father.
    B. John, an officer in the revenue department, came here on Tuesday to meet his ailing father.
    C. John, an officer in the revenue department, have come here on Tuesday to meet his ailing father.
    D. John, an officer in the revenue department, has came here on Tuesday to meet his ailing father.
    E. John, an officer in the revenue department, coming here on Tuesday to meet his ailing father.
19. A. My brothers as well as I are students of this college.
    B. My brothers as well as I am students of this college.
    C. My brothers as well as I was students of this college.
    D. My brothers as well as I has been students of this college.
    E. My brothers as well as I is student of this college.
20. A. Either Hamid or his brother attend the class.
    B. Either Hamid or his brother attends the class.
    C. Either Hamid or his brother are attending the class.
    D. Either Hamid or his brother have attended the class.
    E. Either Hamid or his brother was attend the class.

Choose the most suitable choice (Q. 21-30).

21. A main clause _____.
    A. consists of comma splices
    B. has a subject and a predicate and can stand alone as a sentence
    C. lacks a predicate
    D. has a subject and a predicate, but it cannot stand alone as a sentence
    E. None of these
22. An adverb clause is _____.
    A. a subordinate clause that modifies a verb, an adjective, or an adverb
    B. a subordinate clause that is used as a noun
    C. declarative
    D. interrogative E. None of these
23. A compound sentence _____.
    A. has only one main clause and two predicates
    B. has two or more main clauses
    C. modifies a noun or a pronoun
    D. is a subordinate clause E. None of these
24. Which of the following is a subordinating conjunction?
    A. since D. All of the above
    B. although E. None of the above
    C. when
25. What is true of the following sentence?
    "I don't like him, but he wants me to like him."
    A. It is a simple sentence.
    B. It is a complex sentence.
    C. It is a compound sentence.

D. It is a compound-complex sentence.
E. None of the above.

26. Which part of the following sentence describes a clause?
The internet, (A)/ which has become such a global phenomenon, (B)/ has contributed (C)/ to the predominance of English. (D)/ None of these (E).

27. What is true of the following sentence?
"Mr. Sharma has decided to hire a personal trainer because he is worried about his heart."
A. The underlined section is an independent clause.
B. This sentence has two independent clauses.
C. "Mr. Sharma has decided to hire a personal trainer" is the independent clause.
D. All of the above
E. None of the above

28. "The man from whom she received that recommendation letter works for the government."
The underlined part in the above written sentence functions as-
A. Subject
B. Object
C. Complement
D. Subject predicative
E. Object predicative

29. Which of the following is the best analysis of the given sentence as per basic clause patterns?
"You might hurt yourself by doing like this."
A. Subject – Verb – Indirect Object
B. Subject - Verb - Direct Object
C. Subject - Verb - Indirect Object - Adjunct
D. Subject - Verb - Direct Object - Adjunct
E. Subject - Verb - Adjunct

30. "To win a gold medal in Rio Olympic was her long cherished dream." The underlined portion in this sentence is-
A. Subject
B. Direct object
C. Predicate
D. Adjunct
E. Predicator

Read the passage given below and answer the questions that follow (Q. 31-40):

Sometimes fate is like a small sandstorm that keeps changing directions. You change direction but the sandstorm chases you. You turn again, but the storm adjusts. Over and over you play this out, like some ominous dance with death just before dawn. Why? Because this storm isn't something that blew in from far away, something that has nothing to do with you. This storm is you; something inside of you. So all you can do is give in to it, step right inside the storm, closing your eyes and plugging up your ears so the sand doesn't get in, and walk through it, step by step. There's no sun there, no moon, no direction, no sense of time; Just fine white sand swirling up into the sky like pulverized bones. That's the kind of sandstorm you need to imagine.

And you really will have to make it through that violent, metaphysical, symbolic storm. No matter how metaphysical or symbolic it might be, make no mistake about it: it will cut through flesh like a thousand razor blades. People will bleed there, and you will bleed too; hot, red blood. You'll catch that blood in your hands, your own blood and the blood of others.

And once the storm is over you won't remember how you made it through, how you managed to survive. You won't even be sure, in fact, whether the storm is really over. But one thing is certain. When you come out of the storm you won't be the same person who walked in. That's what this storm's all about." *-Excerpt from 'Kafka on the Shore' by Haruki Murakami,*

31. Find the part of speech of the following expression- 'over and over' in the sentence "Over and over you play this out"
A. Noun
B. Adjective
C. Conjunction
D. Adverb
E. None of the above

32. The word 'pulverized' is used in the text as
A. Verb
B. Adverb
C. Interjection
D. Adjective
E. None of the above

33. The word 'metaphysical' is an adjective form of the noun-
A. Metaphysicalism
B. Metaphysics
C. Metaphysicalization
D. Metaphysicament
E. Metamorphosis

34. "There's no sun **there**"- the emboldened word is actually a-
A. Adverbial of place
B. Object
C. Adjective of place
D. Place noun
E. None of the above

35. "People will bleed there, and you will bleed too"- Identify the clauses in this sentence-
A. Coordinating conjunction
B. Coordinate clauses
C. Subordinate clauses
D. Dependent clauses
E. None of the above

36. "And once the storm is over you won't remember how you made it through"- the underlined part has-
A. One coordinating conjunction
B. One subordinating conjunction
C. Two coordinating conjunctions
D. Two subordinating conjunctions
E. None of the above

37. "Sometimes fate is like a small sandstorm that keeps changing directions"-The underlined clause is actually -
A. An adjectival coordinate clause
B. An adverbial subordinate clause
C. An adjectival relative clause
D. An adverbial relative clause
E. None of the above

38. Find out the problem in the following sentence-" You change (a) direction but (b) the sandstorm chase (c) you unless (d) you wield (e)"-
A. Option (a)
B. Options (b) and (c)
C. Option (c)
D. Option (d)
E. None of the above

39. "No matter how metaphysical or symbolic it might be, make no mistake about it"-what is the subject of the underlined clause?
A. 'No matter'
B. 'it'
C. 'Metaphysical or symbolic'
D. No subject.
E. (You).

40. "So all you can do is give in to it."-Select the correct option.
A. So all you can do are give in to it.
B. So all of you can do is give in to it.
C. So all you can do is given in to them.
D. So you all can do is give in to it.
E. No correction required.

S B

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Odd Semester), Term I Examination, 2016-17

Subject: English for Professional Purposes–II
Time: 40 Minutes

Subject Code: AHE 3083
Max. Marks: 20

VERBAL APTITUDE

Identify the part of speech of the underlined word in Q. 1-10.

1. None except for her could pass the test.
   A. Adjective B. Conjunction C. Preposition
   D. Connector E. None of these
2. She kept her cool in spite of provocations.
   A. Verb B. Adjective C. Noun
   D. Adverb E. None of these
3. The audience was calmed by the leader.
   A. Verb B. Adjective C. Adverb
   D. Noun E. None of these
4. He has lived in India for ten years.
   A. Connector B. Preposition C. Conjunction
   D. Adverb E. None of these
5. The weather is cold outside.
   A. Noun B. Adjective C. Verb
   D. Adverb E. None of these
6. Vikram is rich but Gayatri is poor.
   A. Verb B. Adjective C. Conjunction
   D. Preposition E. None of these
7. I cannot come to office today for I have relatives at home.
   A. Conjunction B. Verb C. Preposition
   D. Adjective E. None of these
8. Fortunately, the baby was saved.
   A. Adjective B. Adverb C. Noun
   D. Verb E. None of these
9. My friends are very cool.
   A. Adverb B. Adjective C. Noun
   D. Verb E. None of these
10. George handled all the questions calmly.
    A. Noun B. Adjective C. Adverb
    D. Verb E. None of these

Identify the part that contains error (Q. 11-15).

11. Getting past that veneer of (A)/ superficial responses was detective work, (B)/ and she simply didn't has the time or the inclination (C)/ to invest so much of herself into the multiple rounds (D)/ of interviews necessary to bring someone aboard (E).
12. The primary caveat, (A)/ however, is to keep (B)/ your questioning patterns (C)/ within legal boundaries so that (D)/ you don't unnecessarily exposed your company (E).
13. Now onto that ever-so-delicate opening icebreaker: (A)/ It's not always easy (B)/ to generate small talk (C)/ before the actual work begin, (D)/ but it always feels natural to do so (E).
14. English was permanently introduce (A)/ into Ireland when the Normans (B)/ invaded the country during the twelfth century (C)/ and settled French and English speakers (D)/ in the eastern coastal region (E).
15. The monograph explore the politics and practices (A)/ of policy-making and policy implementation (B)/ with regard to elementary education (C)/ in India in historical context (D)/ None of these(E).

Identify the grammatically correct sentence (Q. 11-15).

16. A. He is one of the best leaders that have ever lived.
    B. He is one of the best leaders that is ever lived.
    C. He is one of the best leaders that has ever lived.
    D. He is one of the best leaders that was ever lived.
    E. None of these
17. A. Industry, in addition to ability, are needed for success.
    B. Industry, in addition to ability, have needed for success.
    C. Industry, in addition to ability, have been needed for success.
    D. Industry, in addition to ability, is needed for success.
    E. None of these.
18. A. He together with his friends were punished.
    B. He together with his friends were being punished.
    C. He together with his friends was punished.
    D. He together with his friends are punished.
    E. None of these.
19. A. The number of pupils are increasing.
    B. The number of pupils are being increasing.
    C. The number of pupils have been increasing.
    D. The number of pupils is increasing.
    E. None of these.
20. A. The bag, together with the books, are lost.
    B. The bag, together with the books, is lost.
    C. The bag, together with the books, have been lost.
    D. The bag, together with the books, are being lost.
    E. None of these.

Choose the most suitable choice (Q. 21-30).

21. A subordinate clause _____.
    A. is independent D. lacks a verb
    B. contains a main clause E.None of these
    C. has a subject and a predicate, but it cannot stand alone as a sentence
22. An adjective clause is a subordinate clause that _____.
    A. modifies a/an adjective
    B. modifies a/an verb
    C. modifies a/an noun or pronoun
    D. modifies a/an adverb
    E. None of these
23. A complex sentence _____.
    A. has more than one main clause and at least one subordinate clause
    B. has one main clause and one or more subordinate clauses
    C. is a subordinate clause used as a noun
    D. has a single subordinate clause
    E. None of these
24. Which of the following is a coordinating conjunction?
    A. and D. All of the above
    B. for E. None of the above
    C. nor
25. What is true of the following sentence?
    "The person, the most loving one, is here."
    A. It contains a noun clause.
    B. It contains an adverbial clause.
    C. It contains a relative clause.
    D. It contains an appositive phrase.
    E. None of the above.
26. Which part of the following sentence describes a clause?

English is the (A)/ dominant language on the internet because (B)/ the American computer industry dominates (C)/ the rest of the world. (D)/ None of these (E).

27. What is true of the following sentence?
"He is very good as long as his wife is around, but he goes to the freezer for ice-cream when she leaves."
A. This sentence has three independent clauses.
B. This sentence has two independent clauses.
C. This sentence has one independent clause.
D. All of the above
E. None of the above

28. "The man who was speaking to us was very tall."
The underlined part in the above written sentence functions as-
A. Subject
B. Object
C. Complement
D. Subject predicative
E. Object predicative

29. Which of the following is the best analysis of the given sentence as per basic clause patterns?
"College students who spend four years on their degrees make more money than graduates of two-year colleges."
A. Subject – Subject Predicative - Verb – Direct Object - Adjunct
B. Subject - Verb - Direct Object
C. Subject - Verb - Indirect Object - Direct Object
D. Subject - Verb - Direct Object – Adjunct
E. Subject - Verb - Adjunct

30. "That you can make me do the exercises is 100% true, I believe." The underlined portion in this sentence is-
A. Subject
B. Direct object
C. Predicate
D. Adjunct
E. Predicator

Read the passage given below and answer the questions that follow (Q. 31-40):

One of the intriguing aspects of globalising Anglo-American expertise is that ELT is not a high-prestige profession in either the US or the UK. In both countries there are unmet English language needs for children and adults. In addition, foreign language learning is much less widespread and effective than in many countries.

It is true that there is a massive demand for English worldwide, to which many factors, from trade and tourism to regional integration, contribute. Maintaining the value of western investments and influence in the decolonisation period led to the mushrooming of departments of Teaching English as Second Language and applied linguistics from the 1950s. The demand for English has been orchestrated by western governments and their allies worldwide, and key bodies such as the World Bank. The "supply" of expertise dovetails with demand.

Governments have tended to clutch at a quick fix, such as importing native speakers, or starting English ever earlier, either as a subject or as the medium of instruction, in the hope that this will make the learning of English more effective. Such demands should be challenged by ELT when both the demand and the response are unlikely to be educationally, culturally or linguistically well-informed. –

Excerpts from a Guardian report in the Opinion Column, September, 2016.

31. "One of the intriguing aspects of globalising Anglo-American expertise"-the word 'intriguing' is
A. Adverb
B. Adjective
C. Gerund
D. Participle
E. None of the above

32. "One of the intriguing aspects of globalising Anglo-American expertise"-the word 'globalising' is
A. Adjective
B. Adverb
C. Participle Verb
D. Noun
E. None of the above

33. "In addition, foreign language learning is much less widespread.." - In this sentence, what is the status of 'less'?
A. Adverb of degree
B. Conjunction
C. Expletive
D. Adjective of first degree
E. Compound Noun

34. "Maintaining the value of western investments and influence in the decolonisation period led to the mushrooming of departments.."- The words 'maintaining' and 'mushrooming' are-
A. Participle verbs both
B. Progressive tense
C. Participle verb and Gerund
D. Nouns both
E. None of the above

35. "..the response are unlikely to be educationally.." what is the noun form of the word 'unlikely'?
A. Unlike
B. Unlikeability
C. Unlikeliness
D. Unliking
E. None of the above

36. "Such demands should be challenged by ELT when both the demand and the response are unlikely to be educationally, culturally or linguistically well-informed"- What kind of conjunction/connective is 'when' in this sentence?
A. Adverbial
B. Coordinating
C. Nominal
D. Matrix
E. None of the above

37. Find the incorrect part of the sentence-"In both countries (a) there are (b) unmet English language needs (c) for children and adults (d) which has to be addressed (e) soon."
A. Part (a)
B. Part (b)
C. Part (c)
D. Part (d)
E. Part (e)

38. Find the incorrect part of the sentence-"In addition, (a) foreign language learning classes (b) is much(c) less widespread (c) and effective than (d) in many countries (e)."
A. Part (a)
B. Part (b)
C. Part (c)
D. Part (d)
E. Part (e)

39. "Governments have tended to clutch at a quick fix.."- The word 'to cutch' is
A. A verb
B. A preposition
C. A noun
D. An adverb
E. An adjective

40. "The demand for English has been orchestrated by western governments and their allies worldwide."- Identify the coordinating conjunction in this sentence.
A. The
B. And
C. Their
D. Orchestrated
E. None of the above

SET-C

University Roll No._____

GLA UNIVERSITY, MATHURA

B. Tech. 2nd Year (Odd Semester), Term I Examination, 2016-17

Subject: English for Professional Purposes–II | Subject Code: AHE 3083

Time: 40 Minutes | Max. Marks: 20

VERBAL APTITUDE

Identify the part of speech of the underlined word in Q. 1-10.

1. None <u>but</u> a technician can fix the AC.
   A. Connector B. Preposition C. Adjective
   D. Conjunction. E. None of these
2. Vikram couldn't play the match <u>for</u> he injured his knee.
   A. Preposition B. Adjective C. Conjunction
   D. Verb E. None of these
3. Great achievers are often from a <u>humble</u> background.
   A. Noun B. Verb C. Adjective
   D. Adverb E. None of these
4. His writing is good <u>but</u> not his spelling.
   A. Conjunction B. Adjective C. Preposition
   D. Verb E. None of these
5. Age hasn't <u>greyed</u> her hair.
   A. Adjective B. Verb C. Noun
   D. Adverb E. None of these
6. This is a train <u>for</u> Jhansi.
   A. Conjunction B. Connector C. Adverb
   D. Preposition E. None of these
7. <u>Obviously</u> she is innocent.
   A. Adjective B. Adverb C. Noun
   D. Verb E. None of these
8. Sanjana's mother has lots of <u>grey</u> hair.
   A. Adverb B. Noun C. Adjective
   D. Verb E. None of these
9. She has caught a <u>cold</u>.
   A. Adjective B. Adverb C. Noun
   D. Verb E. None of these
10. Difficulties in life have <u>humbled</u> him.
    A. Adjective B. Adverb C. Verb
    D. Noun E. None of these

Identify the part that contains error (Q. 11-15).

11. In the event that the Purchaser default in the payment (A) / of any installment of purchase price, taxes, insurance, interest, or the annual charge (B) / described elsewhere herein, (C) / or shall default in the performance of (D) / any other obligations set forth in this Contract (E).
12. One of the reasons why children from unlettered homes (A) / was at a disadvantage when they start learning to read (B) / may be that they lack the familiarity (C) / with the shapes of words and letters(D) / None of these(E).
13. You mean (A) / to say that (B) / either he or his brother (C) / have stolen that book (D) / None of these (E).
14. If your conversation leads to some kind of reasonable compromise (A) / where the candidate agree that (B) / it would be reasonable to ask a subordinate to leave the hardware at home (C) / or to take it off whenever dealing with customers (D) / None of these (E).
15. I, being young (A) and you, old and experienced, (B) / were afraid of (C) / your age and dignity. (D) / None of these (E).

Identify the grammatically correct sentence (Q. 16-20).

16. A. Four hundred rupees are not a big amount for you.
    B. Four hundred rupees have not been a big amount for you.
    C. Four hundred rupees were not a big amount for you.
    D. Four hundred rupees is not a big amount for you.
    E. Four hundred rupees had not a big amount for you.
17. A. The manager and Principal of the school are absent.
    B. The manager and Principal of the school is absent.
    C. The manager and Principal of the school were absent.
    D. The manager and Principal of the school have absent.
    E. The manager and Principal of the school are being absent.
18. A. Neither he nor his friends are present in the school yesterday.
    B. Neither he nor his friends is present in the school yesterday.
    C. Neither he nor his friends has been present in the school yesterday.
    D. Neither he nor his friends were present in the school yesterday.
    E. Neither he nor his friends be present in the school yesterday.
19. A. If you take exercise regularly, you can keep fit and does a lot of work.
    B. If you takes exercise regularly, you can keep fit and do a lot of work.
    C. If you take exercise regularly, you can keep fit and do a lot of work.
    D. If you take exercise regularly, you can keep fit and doing a lot of work.
    E. If you taken exercise regularly, you can keep fit and done a lot of work.
20. A. It looks like you attends De Paul University in Chicago and graduated in 2005.
    B. It look like you attended De Paul University in Chicago and graduated in 2005.
    C. It looks like you attends De Paul University in Chicago and graduate in 2005.
    D. It looks like you attended De Paul University in Chicago and graduated in 2005.
    E. It looks like you attended De Paul University in Chicago and graduats in 2005.

Choose the most suitable choice (Q. 21-30).

21. A coordinate clause _____.
    A. is independent D. contains a main clause
    B. lacks a verb E. None of these
    C. has a subject and a predicate, but it cannot stand alone as a sentence
22. A noun clause _____.
    A. modifies a verb
    B. modifies a pronoun
    C. is a subordinate clause used as a verb
    D. is a subordinate clause used as a noun
    E. None of these
23. A simple sentence _____.
    A. has more than one main clause and at least one subordinate clause
    B. has one main clause and one or more subordinate clauses
    C. is a subordinate clause used as a noun
    D. has a single subordinate clause
    E. has a single main clause
24. Which of the following functions both as a conjunction and as an adverb?
    A. nevertheless D. All of the above
    B. however E. None of the above
    C. thus
25. What is true of the following sentence?
    "Many people probably think that politeness is simply a social lubricant."

A. It contains a prepositional phrase.
B. It contains an adjectival clause.
C. It contains an adverbial clause.
D. It contains a noun clause.
E. None of the above.

26. Which part best describes a clause?
My friend, (A)/ who teaches English in Yeman, (B)/ says that she (C)/ had more students eight years ago. (D)/ None of these (E).

27. What is true of the following sentence?
"Dr. Ghosh must learn that eating all those sweets may give him a temporary pleasure but that it's not good for his heart and that he would feel better about himself if he stopped eating all those rich and sweet foods that are not good for him."
A. "Dr. Ghosh must learn" is a good example of a dependent clause.
B. The independent clause begins with the first "that."
C. This sentence has several independent clauses.
D. All of the above
E. None of the above

28. "Do you know the name of the artist who painted this portrait?"
The underlined part in the above written sentence functions as-
A. Subject
B. Direct Object
C. Indirect Object
D. Object predicative
E. None of these

29. Which of the following is the best analysis of the given sentence as per basic clause patterns?
"She believes that American families used to take more pride in learning second languages."
A. Subject – Verb – Indirect Object
B. Subject - Verb - Direct Object
C. Subject - Verb - Indirect Object - Direct Object
D. Subject - Verb - Indirect Object – Adjunct
E. Subject - Verb – Adjunct

30. "Do or Die has become the motto of final year students for they want to be placed in Infosys." The underlined portion is-
A. Subject
B. Direct object
C. Predicate
D. Adjunct
E. Predicator

Read the passage given below and answer the questions that follow (Q. 31-40):

We are therefore in 1880 dealing not so much with a single world, as with two sectors combined together into one global system: the developed and the lagging, the dominant and the dependent, the rich and the poor. Even this description is misleading. While the (smaller) first world, in spite of its considerable internal disparities, was united by history and as the common bearer of capitalist development, the (much larger) second world was united by nothing except its relations with, that is to say its potential or actual dependency on, the first. What else, except a common membership of the human race, had the Chinese Empire in common with Senegal, Brazil with the New Hebrides, Morocco with Nicaragua? The second world was united by neither history, culture, social structure nor institutions, nor even by what we today think of as the most salient characteristic of the dependent world, namely mass poverty. For wealth and poverty as social categories apply only to societies stratified in a certain way, and to economies structured in a certain way, and parts of the dependent world were, as yet, neither. All human societies known to history contain some social inequalities (apart from those between the sexes), but if Indian maharajahs visiting the west could be treated as though they were millionaires in the western sense, the big men or chiefs in New Guinea could not be so assimilated, even notionally. – *Excerpt from The Age of Empire by Eric Hobsbawm.*

31. "with two sectors combined together into one global system.."-What is the status of the word 'into' here-
A. Conjunction
B. Adjunct
C. Preposition
D. Pronoun
E. Adjectival complement

32. "in spite of its considerable internal disparities.."-What is the part of speech of the word 'in spite'?
A. Adverb
B. Preposition
C. Adjective
D. Conjunction
E. None of the above

33. "Even this description is misleading"- the word 'misleading' here is a
A. Gerund
B. Adjective
C. Participle
D. Noun
E. None of the above

34. Find the erroneous part in this sentence-"All human (a) societies (b) known to (c) history contain (d) some social inequalitiy (e)."
A. Part (a)
B. Part (b)
C. Part (c)
D. Part (d)
E. Part (e)

35. "the big men or chiefs in New Guinea could not be so assimilated, even notionally"-The word 'notionally' is functioning in this sentence as-
A. Adverb
B. Adjective
C. Object
D. Predicate
E. Noun

36. "While the (smaller) first world, in spite of its considerable internal disparities, was united by history and as the common bearer of capitalist development, the (much larger) second world was united by nothing except its relations with, that is to say its potential or actual dependency on, the first."-The underlined is an example of –
A. Coordinate conjunction
B. Subordinate conjunction
C. Coordinate clause
D. Subjunctive clause
E. None of the above

37. "For wealth and poverty (a) as social categories (b) applies only (c) to societies stratified (d) in a certain way (e)"-
A. Part (a)
B. Part (b)
C. Part (c)
D. Part (d)
E. Part (e)

38. "if Indian maharajahs visiting the west could be treated as though they were millionaires in the western sense, the big men or chiefs in New Guinea could not be so assimilated, even notionally"-The underlined word is a –
A. Subordinating conjunction
B. Preposition
C. Adverbial Adjunct
D. Independent Clause
E. None of the above

39. "as social categories apply only to societies stratified in a certain way.."-The word 'stratified' is actually a
A. Verb
B. Nominal clause
C. Adjective
D. Demonstrative adverb
E. None of the above

40. In the expression – "One global system: the developed and the lagging" – the word "lagging stands as -
A. Adjective
B. Gerund
C. Noun
D. Progressive tense
E. None of the above